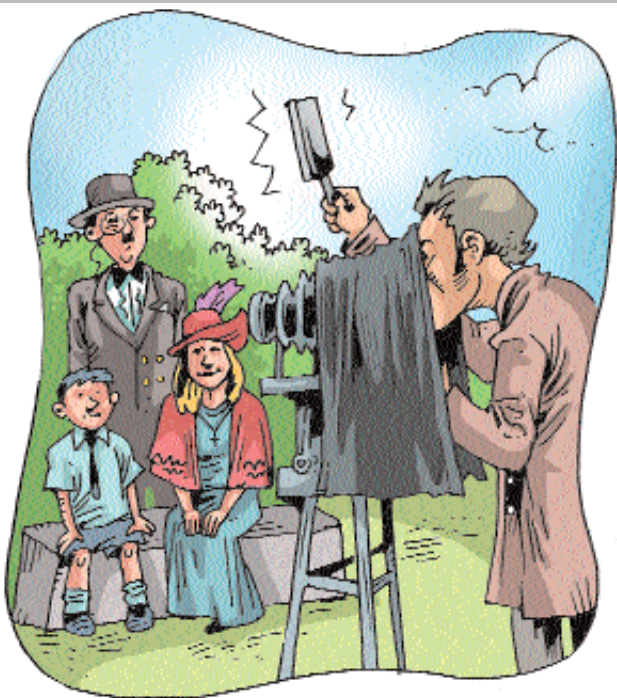


DIGITAL CAMERAS



Shooting In RAW Format

If your camera has a RAW capture mode, it's a good idea to try it out. Here are some things you need to know.

What is RAW?

RAW is not really a single file format at all. Every camera manufacturer has its own native file format; all these formats are collectively referred to as RAW. For example, Canon RAW files use the .crw and .cr2 extensions, and Nikon cameras use the .nef extension. They are all unique as well proprietary—the files in these various formats are incompatible.

But no matter what flavour of RAW you use, the basic idea is the same. RAW files are uncompressed, perfect representations of what the camera's image sensor captures when you press the shutter release. RAW images are totally unprocessed. Contrast this with JPEG photos, where the camera performs tasks such as white balance adjustment, colour correction, and image sharpening.

Most importantly, RAW files preserve all the colours

originally captured by the camera, usually 12 bits per pixel. When a picture is saved as a JPEG, the camera reduces the total number of colours to 8 bits per pixel. The higher colour fidelity of RAW makes for better photos, especially if you do a lot of editing on your images.

Disadvantages Of RAW

There are a couple of disadvantages to working with RAW, however. First and foremost is the time it takes to save each image on your camera. Most modern cameras can save even large

JPEG images more or less as fast as you can shoot them. But RAW images are another story. On a typical 6-megapixel camera, for instance, each RAW image takes about 40 seconds to get saved to the memory card.

File size is another consideration. Top-quality JPEG images could typically be just above 2 MB each, but their RAW counterparts are about 4 to 5 MB. If you have a smaller memory card, this is something to consider.

Fortunately, there's nothing particularly unusual about moving RAW files from the camera to the PC. Use whatever technique you prefer. But after the pictures are on your computer, RAW is a different story.

RAW-Compatible Software

In the past, most image editors didn't understand the RAW format, and it was also difficult to view, edit, and organise RAW photos. But thankfully, that's changed. No matter what popular commercial photo editor you use, you'll get RAW support if you upgrade to the newest version.

However, Windows XP doesn't understand RAW files. In your folders, RAW images appear as nondescript icons instead of thumbnail previews. And if you double-click on a RAW file, it won't open the Picture and Fax Viewer to display the image.

There is, however, a

solution: Microsoft's free RAW Image Thumbnailer and Viewer. This PowerToy lets you preview, see thumbnails, and print RAW images as if they were any other file format that Windows understands. It also gives you the ability to see RAW images on the desktop without special RAW image management software. Download the PowerToy for XP from www.microsoft.com/windowsxp/downloads/powertoys/xppowertoys.msp.

Depending upon what image editor you use to edit RAW files, your experience will vary. Most programs allow you to make some initial adjustments to your RAW photo before loading it onto the editing canvas—more on this later.

Should You Use RAW?

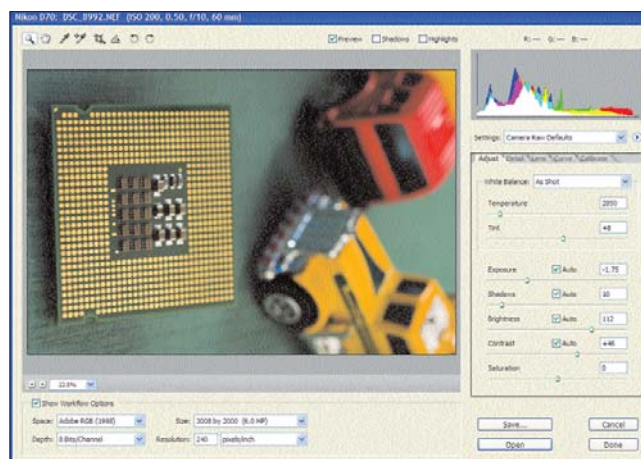
If you need to spend money to experiment with RAW, think twice. JPEG, used with low compression, gives results that are surprisingly similar to RAW. RAW's real benefit comes in by making it easier to get high-quality results when you edit your photos on the PC. If you rarely spend much time editing your photos, using RAW may be more trouble than it's worth.

RAW Gives Richer Pictures

With a RAW file, the process of applying the required algorithms is given to your computer rather than to your camera. This means the conversion process can take advantage of the more powerful processor in your computer as opposed to the processor in your camera. And, because your computer's more powerful processor is being used, camera manufacturers have the option of using more powerful algorithms to process your photograph than could be used in the camera. This can have the effect of giving you richer, sharper pictures.

Pre-Edit In RAW

A great reason for shooting in RAW is the ability to make exposure, white



If you shoot in the RAW format, you will find myriad options by which you can edit your photographs